WORKSHOP

BPMS & Camunda 8

AI-driven Process Analysis & Re-Design

BPM – SoSe 2026 | Rafael Barros, Daniel Kern, Manuel Stamm, Yusuf Aslan

Intro & Motivation | BPMS & Camunda 8 | AI Trends | Live Demo: AI-Driven Vendor selection | Showcase 1 | Showcase 2 | Architecture & Roadmap | Q&A

01

Intro & Motivation

Why Business Processes Matter

1/3

of work time is spent on non-value-adding activities

McKinsey, 2022

66%

of organizations have automated at least one business process

McKinsey, Feb 2024

30%

operational cost reduction with hyperautomation + process redesign

Gartner, 2021

Business processes are the backbone of every organization. Without systematic management, organizations lose time, quality, and money through hidden inefficiencies.

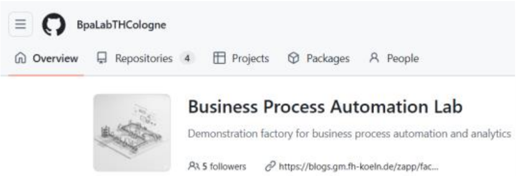
Process Identification — **Process Discovery** — **Process Analysis** — **Process Redesign** — **Process Implementation** — **Process Monitoring**

BPM Lifecycle (Dumas et al., 2018)

Sources: McKinsey, "How good are your internal operations — really?" (mckinsey.com, 2022) | McKinsey & Co., State of Business Automation, Feb 2024 | Gartner Press Release, "Hyperautomation-Enabling Software Market", Apr 2021 (gartner.com)

Our Context: The BPA Lab Bicycle Factory

Virtual Bicycle Manufacturer



A fictive end-to-end bicycle production and shipping company implemented at TH Köln. Built on Camunda 8 BPMS, it covers the complete order-to-delivery value chain.

6 core processes

Order Management → Warehouse → Production Control → Purchasing → Manufacturing → Shipment

(Potential Candidate for AI use-cases) - Purchasing

Our project focus: Purchasing Process → AI-driven analysis & redesign

Order Mgmt

Production Control

Purchasing

Manufacturing

Shipment

Warehouse

02

BPM & Camunda 8

Foundations, landscape, Zeebe architecture, BPA Lab integration

What is a Business Process Management System?

***A BPMS** is software that supports the design, automated execution, monitoring, and analysis of business processes — using BPMN models as the executable specification.*

Model

Design processes visually using BPMN 2.0 notation — gateways, tasks, events, message flows

Execute

Deploy and run BPMN models as live processes orchestrating humans, services & systems

Monitor

Real-time dashboards for SLA tracking, bottleneck detection, and compliance auditing

Optimize

Feed insights back into redesign using process mining, KPIs, and AI-generated suggestions

Why is BPMS relevant? For What? Processes are often not transparent, manually executed, full of leakages.

Camunda 8 is the BPMS used in the BPA Lab — open-source, cloud-native, with built-in connectors and AI integration support.

Sources: Dumas, La Rosa, Mendling & Reijers (2018), Fundamentals of Business Process Management, 2nd ed., Springer · DOI 10.1007/978-3-662-56509-4 | OMG (2014), BPMN 2.0.2 specification (ISO/IEC 19510) · omg.org/spec/BPMN/2.0.2

BPMS in the Process-Aware Systems Landscape

Process-Aware Information Systems (PAIS) — where does BPMS fit?

Domain-specific	Domain-independent (applicable across processes & domains)		
ERP / Ticketing	**BPMS**	**iPaaS**	**RPA**
SAP, Jira	*Camunda 8*	*Make, Zapier*	*UiPath, Blue Prism*
Automates specific domains (finance, ITSM)	General-purpose process orchestration via BPMN	Cloud integration & connector workflows	Automates GUI-level routine tasks
	★ **our focus**		

Key positioning point

A BPMS is the **orchestrator** — it sits above ERP, iPaaS, and RPA tools and coordinates them through BPMN models. It is the only PAIS category whose core artifact (the process model) is both the design language *and* the runtime behavior.

Sources: Dumas et al. (2018), Fundamentals of BPM, 2nd ed. — Ch. 9 "Process Automation" · springer.com/book/978-3-662-56509-4 | Zapp, M. (2025), T3 Trends in BPA — TH Köln BPM Module

From BPMS Theory to Practice: Camunda 8 & Zeebe

Camunda 8 is the BPMS we use in the BPA Lab — here's its technical architecture:

Clients

- Modeler (BPMN design)
- Tasklist (human tasks)
- Operate (monitoring)
- Custom App (Job Workers)

gRPC / REST

Architecture, how camunda processes, interface

Zeebe Gateway · stateless, sessionless entry point — load-balanced

Zeebe Brokers (cluster)

Peer-to-peer · event-sourced · partitioned · no central database
Executes BPMN, manages state, dispatches jobs to workers

Exporter → Elasticsearch

audit log, history, KPIs

What makes Zeebe different from a traditional BPMS engine

No central DB: Event-sourced; brokers replicate state peer-to-peer

Horizontal scaling: Add brokers → throughput scales linearly

External job workers: Business logic decoupled from engine (any language)

Sources: Camunda Docs — Zeebe Architecture · docs.camunda.io/docs/components/zeebe/technical-concepts/architecture/ | Camunda Platform Overview · camunda.com/platform/zeebe/ | Camunda GitHub · github.com/camunda/camunda

BPMN 2.0 – Key Elements in the BPA Lab

Start / Intermediate / End Event

Triggers and terminates process instances

User Task

Human action required (e.g. vendor selection in Purchasing)

Service Task

Automated job worker call (e.g. check stock, send MQTT)

Message Event

Inter-process communication (e.g. Order → Production)

Gateway (XOR/AND)

Branch: parallel tasks or exclusive condition check

Sub-Process

Encapsulated flow, e.g. Manufacturing inside Production

Source: OMG (2014), BPMN 2.0.2 specification · omg.org/spec/BPMN/2.0.2 (formally ISO/IEC 19510)

Camunda 8 in the BPA Lab — and Where AI Extends It

What runs in the BPA Lab today

Camunda 8 Self-Managed

Zeebe broker, Operate, Tasklist, Elasticsearch — all in Docker

6 BPMN process apps

Order Mgmt · Warehouse · Production · Purchasing · Manufacturing · Shipment

Job Workers (Java / Node.js)

Subscribe to Zeebe via gRPC, execute service tasks

MQTT bridge

Connects engine to the physical Fischertechnik factory model

Where AI extends this architecture

Purchasing BPMN — Service Task

`"recommend-vendor"`

AI Job Worker (new)

subscribes to Zeebe via gRPC

LLM API (Claude / GPT-4)

vendor scoring, recommendation

↑ result returned as job variables → next BPMN step

Camunda 8's external job worker pattern makes AI integration clean: the LLM is just another worker, decoupled from the engine.

Bridge to next section: We've now seen what a BPMS is, where Camunda 8 fits, and how AI plugs in. Next → AI Trends in BPM.

Sources: BPA Lab Repository · github.com/BpaLabTHCologne/bpa_lab_demonstration_factory | Camunda Docs — Job Workers · docs.camunda.io/docs/components/concepts/job-workers/

Camunda 8 – Architecture & Components

UI Layer	Tasklist (human tasks)	Operate (monitoring)	Optimize (analytics)
Engine Layer	Zeebe (BPMN engine)	Gateway (gRPC/REST)	Connectors (pre-built integrations)
Data Layer	Elasticsearch (audit log)	MySQL / external DBs	MQTT (IoT events)

BPA Lab uses self-managed Camunda 8 running in Docker — Zeebe engine + Tasklist + Operate + Elasticsearch + custom Job Workers (Java/Node.js)

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AI / Trending Topic

Generative AI in Business Process Analysis & Re-Design

Intelligent Automation: Where AI meets BPM

a) AI Services inside automated processes

User Task — Service Call — AI Service (e.g. OCR, classification) — Auto Decision — Service Call — User Task

b) AI Agents executing sub-processes autonomously (e.g. LLM-driven)

Traditional Process (rule-based steps)

- Static, predefined workflow
- Requires every edge-case to be coded
- Breaks on exceptions

AI Agent Sub-Process (LLM + tools)

- Perceives environment, reasons, acts
- Handles ambiguous inputs & exceptions
- Calls tools, APIs, databases autonomously

Our Project: GenAI (GPT-4 / Claude) as assistant to process analysts → analysing as-is, identifying waste, proposing redesign for the Purchasing Process

Generative AI in Process Analysis & Re-Design

Process Description → BPMN Draft

Feed natural-language process description; GenAI generates BPMN XML or Mermaid flow

Weakness Identification

Prompt AI with as-is process; it applies heuristics (e.g. elimination, automation) to flag waste

Redesign Suggestions

AI proposes to-be variants, explains trade-offs; analyst validates and selects

Vendor & Decision Support

AI scores vendor options (price, lead time, reliability) and recommends best choice in Purchasing

Key Limitation: GenAI lacks deep domain context, may hallucinate process facts, raises data privacy concerns → human-in-the-loop is essential

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Live Demo – AI-Driven Vendor Selection and Analysis

Camunda Tasklist + BPA Lab Order Flow

Live Demo – What We'll Show

1 **Open Camunda Tasklist**

Navigate to the running BPA Lab Camunda 8 instance. Show Operate dashboard — running process instances.

2 **Start an Order**

Submit a new customer order (Choose Bike model, quantity). Observe process instance created in Operate.

3 **Stock Check Fails → Purchasing**

Component stock insufficient — Purchasing process is triggered. Show BPMN gateway routing.

4 **Vendor Selection (User Task)**

Human task: select vendor from list. Discuss: this is the step we target for AI-driven automation (See Demo).

5 **Order Completed**

Purchase order confirmed, stock updated, manufacturing triggered. Full end-to-end trace visible in Operate.

💡 *Key observation: step 4 (vendor selection) is currently manual and slow — this is the redesign target for our AI showcase.*

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Showcase 1

AI-assisted Process Analysis: Identifying Weaknesses in the Purchasing Process

Showcase 1 – As-Is Analysis of the Purchasing Process

As-Is: Purchasing Process Steps

1. Stock shortage detected by Production Control
2. Purchase process triggered via message
3. **User selects vendor manually (chooseVendor form)**
4. Purchase order created in database
5. Bike component quantity increased in stock
6. Production continues / Manufacturing triggered

AI-Identified Weaknesses (Prompt Example)

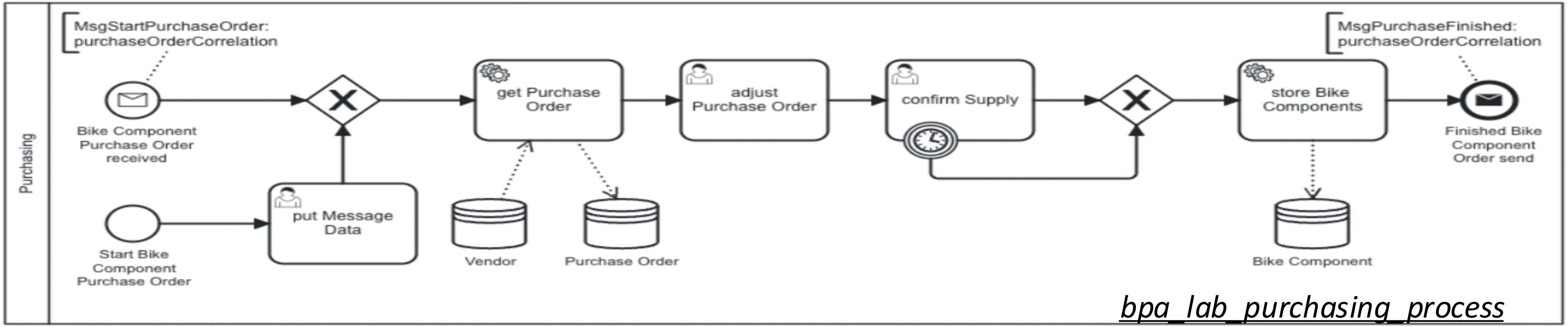
"Given this BPMN process description of the Purchasing process, identify inefficiencies and suggest improvements using process heuristics."

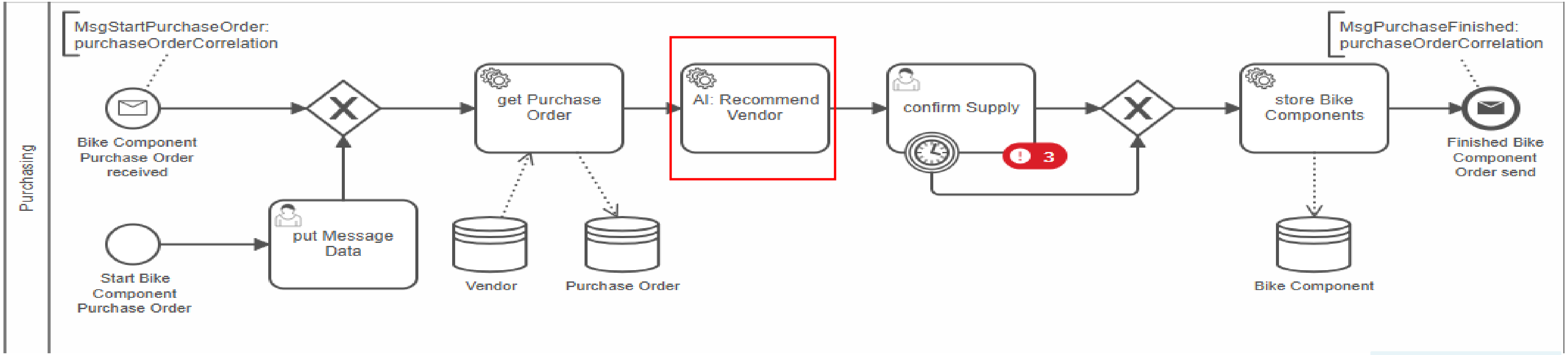
High	**Manual vendor selection**	Waiting time / human bottleneck
High	**Reactive reordering only**	No predictive demand planning
Med	**No vendor scoring**	Sub-optimal cost & quality decisions
Med	**Single vendor per order**	No split-order optimization

→ Biggest finding: Step 3 (vendor selection) is a pure manual task — prime AI target

Sources: Agentic BPM - https://link.springer.com/chapter/10.1007/978-3-032-02936-2_3

Showcase 1 – As-Is Analysis of the Purchasing Process





More in DEMO..

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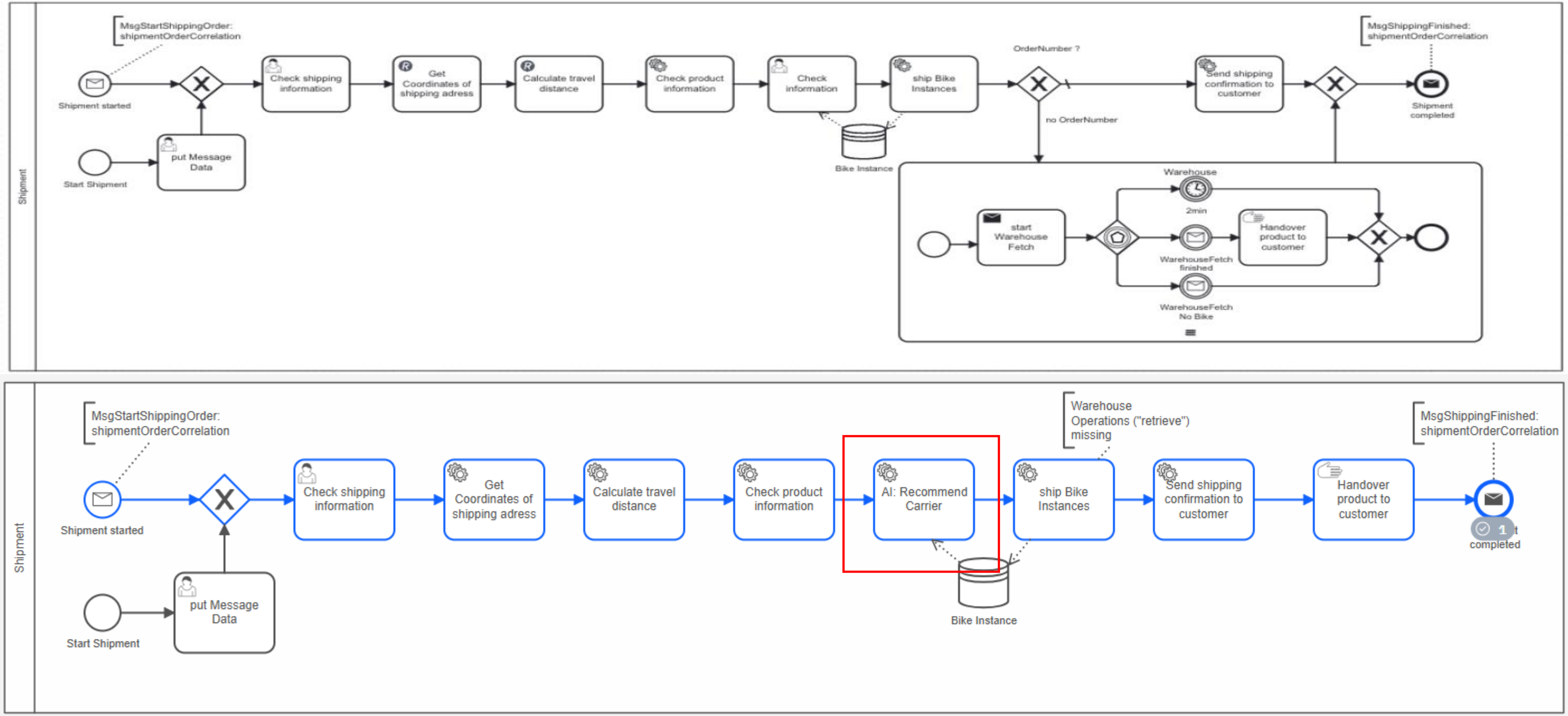
Showcase 2

AI-assisted Process Re-Design: Redesigning the Shipping Process with GenAI

Showcase 2 – As-Is Analysis of the Shipment Process



More in DEMO..

Showcase 2 – To-Be: AI-assisted Shipping Redesign

BEFORE (As-Is)		AFTER (To-Be with AI)
Manual carrier selection from a static form	→	Multi-criteria carrier scoring (price · transit · reliability · CO_2) returns a ranked list
One scoring rule for all shipments	→	Scoring weights shift by urgency and sustainability mode
Human checks country/weight feasibility by hand	→	Infeasible carriers auto-filtered with rationale (weight, country, deadline)
Every order routed through human review	→	Human approves only flagged exceptions — routine orders go straight through

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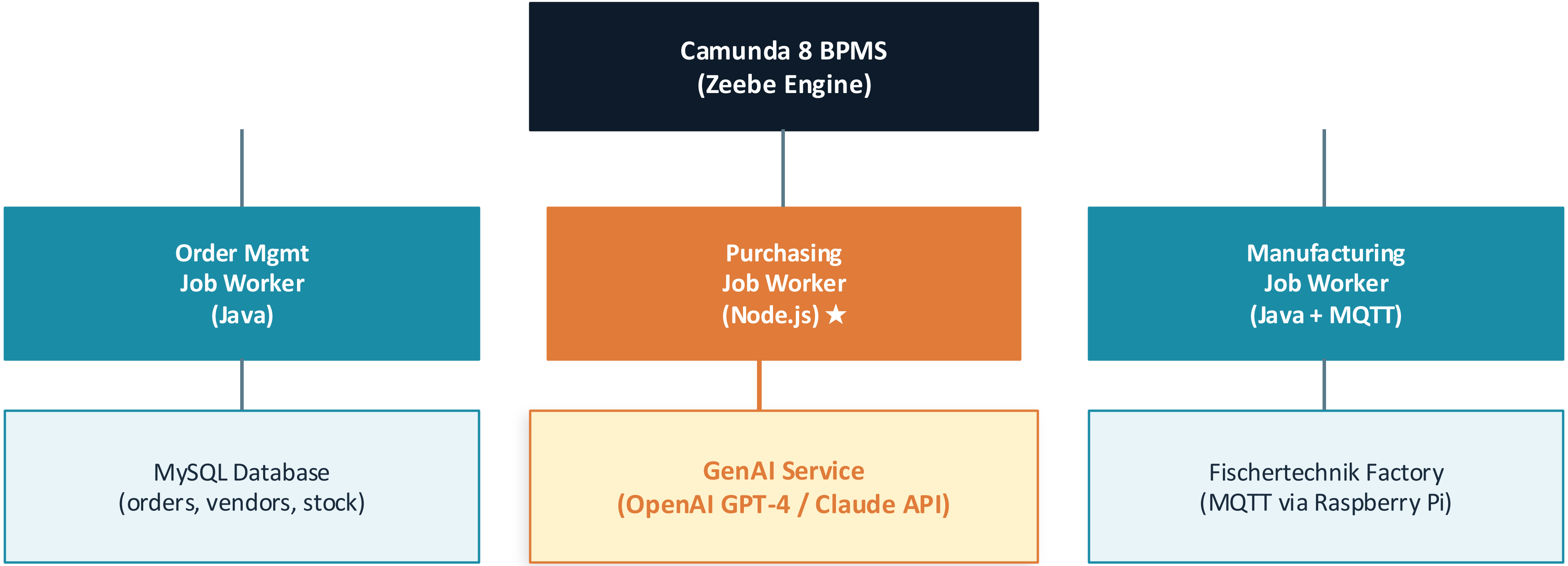
AI Integration Architecture & Roadmap

Solution architecture · data flow & privacy · limitations & best practices · project roadmap

Architecture – AI Integration into Camunda 8



★ **Purchasing Job Worker is our AI integration target:** A new AI Service Task calls the GenAI API with vendor data and order context. The LLM scores and ranks vendors, returns a structured recommendation. A human user task handles final approval only for high-value or exceptional orders.

Sources: BPA Lab Repository · github.com/BpaLabTHCologne/bpa_lab_demonstration_factory | Camunda Docs — Job Workers · docs.camunda.io/docs/components/concepts/job-workers/

Data Flow & Privacy by Design

What data flows in, what comes back — and where privacy is enforced

1 · INPUT to LLM

Sent to the model:

- Component & quantity
- Vendors: price, lead time, reliability
- Order deadline
- Approval threshold rules

Pseudonymized: vendor IDs, no names / PII

2 · LLM

GPT-4 / Claude

Scores each vendor on weighted criteria

Generates ranked recommendation + reasoning

3 · OUTPUT

Returned as job variables:

- Recommended vendor ID
- Confidence / score
- Human-readable reasoning
- Routing flag (auto vs. review)

→ Zeebe resumes the BPMN flow at the next step

Privacy & control safeguards (best practice)

🔒 Data minimization — only decision-relevant fields leave the system

🆔 Pseudonymization — vendor IDs, no names or customer PII

👤 Human-in-the-loop — buyer confirms exceptions & high-value orders

Sources: Camunda Docs — Job Workers (variable passing) · docs.camunda.io/docs/components/concepts/job-workers/ | EU GDPR Art. 5 (data minimization, purpose limitation) · gdpr-info.eu/art-5-gdpr/

Limitations & Best Practices

Three limitations the project brief asks us to address — and how we mitigate each

Limitation	Risk	Mitigation
Domain-specific context	**Risk:** LLM lacks knowledge of factory-specific rules, contracts, or unwritten supplier relationships	→ **Mitigation:** Provide context in the prompt (process description, vendor history, business rules); use retrieval from the MySQL DB
Data privacy	**Risk:** Vendor terms, pricing, and order data are commercially sensitive; sending to a 3rd-party API is a risk	→ **Mitigation:** Data minimization + pseudonymization; consider self-hosted / EU-region models for production use
Decision-making accuracy	**Risk:** LLMs can hallucinate facts or produce confident-but-wrong recommendations	→ **Mitigation:** Human-in-the-loop for exceptions; AI explains its reasoning; validate against ground-truth KPIs

Guiding principle: AI is an assistant to the process analyst, not a replacement — it accelerates analysis, the human owns the decision.

Sources: BPM2026 Project Brief — "AI-driven Process Analysis & Re-Design" (TH Köln) | van Dun et al. (2023), ProcessGAN, Decision Support Systems · doi.org/10.1016/j.dss.2022.113880

Project Roadmap – Three Phases

Phase 1 · ✓ NOW

Research & Workshop

- Literature review (Dumas et al. + ProcessGAN paper)
- Document as-is Purchasing BPMN
- Identify AI integration points
- Deliverable: this Workshop (40 min)

Phase 2 · → NEXT

First Showcase

- Build GenAI prompt for process analysis
- Draft to-be BPMN with AI service task
- Manual prompting demonstration
- Deliverable: Intermediate Presentation

Phase 3 · FINAL

Final Showcase & Assessment

- Refine showcase (API integration or advanced prompting)
- Assess limitations & best practices
- Client-facing consulting perspective
- Deliverable: Final Presentation + Expert Talk

Q & A

Questions & Discussion

- Where would YOU trust AI to decide — and where would you always want a human?
- What risks do you see in using GenAI for vendor selection in a real company?
- How could process mining data improve AI-generated redesign suggestions?